

Algebraic K-Theory and Topological Hochschild Homology

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Status. These notes retain the basic infinity-category and Tate-construction arguments and the compactly generated duality calculation. The continuous K-theory, precompact ideals and rigidity programme remains in supporting pending a complete hypotheses and proof audit.

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Chapter 1

Preliminaries

In this note, all categories are $(\infty, 1)$ -categories.

1.1 ∞ -Categories

Definition 1.1.1. A quasi-category or an ∞ -category $\mathcal{C}$ is a weak Kan complex, i.e. it is a simplicial set such that for all $0 < i < n$, every map of simplicial sets $\sigma_0: \Lambda_i^n \rightarrow \mathcal{C}$ can be extended to a map $\sigma: \Delta^n \rightarrow \mathcal{C}$.

Example 1.1.2. Every Kan complex is an ∞ -category, we will denote the ∞ -category of Kan complex as $\mathcal{S}$.

For every ordinary category $\mathcal{C}$, its nerve $N_\bullet(\mathcal{C})$ is an ∞ -category. We will abuse terminology by identifying an ordinary category with its nerve.

1.2 (Un-)straightening equivalence

Theorem 1.2.1. For any ∞ -category $\mathcal{C}$, there exists an equivalence given by straightening/unstraightening

$$\mathrm{coCart}(\mathcal{C}) \simeq \mathrm{Fun}(\mathcal{C}, \mathrm{Cat}_\infty)$$

where $\mathrm{coCart}(\mathcal{C})$ is the category of coCartesian fibrations over $\mathcal{C}$.

1.3 Localization of ∞ -Categories

Theorem 1.3.1. *Let $\mathcal{C}$ be a small, stable ∞ -category and $\mathcal{D} \subset \mathcal{C}$ a stable subcategory.*

1. *Let W be the collection of all arrows in $\mathcal{C}$ whose cofibre lies in $\mathcal{D}$, then the Dwyer-Kan localization $\mathcal{C}/\mathcal{D} := \mathcal{C}[W^{-1}]$ is a stable ∞ -category, and $\mathcal{C} \rightarrow \mathcal{C}/\mathcal{D}$ is an exact functor.*

For any stable ∞ -category $\mathcal{E}$, the functor $\mathcal{C} \rightarrow \mathcal{C}/\mathcal{D}$ induces an equivalence between $\mathrm{Fun}^{ex}(\mathcal{C}/\mathcal{D}, \mathcal{E})$ and the full subcategory of $\mathrm{Fun}^{ex}(\mathcal{C}, \mathcal{E})$ consisting of those functors which send all objects of $\mathcal{D}$ to 0.

2. *Let $X, Y \in \mathcal{C}$ with image $\bar{X}, \bar{Y} \in \mathcal{C}/\mathcal{D}$. The mapping space in $\mathcal{C}/\mathcal{D}$ is given by*

$$\mathrm{Map}_{\mathcal{C}/\mathcal{D}}(\bar{X}, \bar{Y}) \simeq \mathrm{colim}_{Z \in \mathcal{D}/Y} \mathrm{Map}_{\mathcal{C}}(X, \mathrm{cofib}(Z \rightarrow Y)),$$

where the colimit is filtered.

3. *Assume that $\mathcal{E}$ is a presentable stable ∞ -category. Then the inclusion $\mathrm{Fun}^{ex}(\mathcal{C}/\mathcal{D}, \mathcal{E}) \subset \mathrm{Fun}^{ex}(\mathcal{C}, \mathcal{E})$ is right adjoint to a localization.*

Proof. (2): Consider the inclusion $\mathrm{Fun}(\mathcal{C}[W^{-1}], \mathcal{S}) \subset \mathrm{Fun}(\mathcal{C}, \mathcal{S})$, then it admits a left adjoint given by localization functor

$$L: \mathrm{Fun}(\mathcal{C}, \mathcal{S}) \rightarrow \mathrm{Fun}(\mathcal{C}[W^{-1}], \mathcal{S}).$$

Lemma 1.3.2. *Let $X \in \mathcal{C}$ with image $\bar{X} \in \mathcal{C}[W^{-1}]$, localization gives the functor*

$$\mathrm{Map}_{\mathcal{C}[W^{-1}]}(\bar{X}, -) \simeq L(\mathrm{Map}_{\mathcal{C}}(X, -)).$$

Moreover, the following diagram

$$\begin{array}{ccc} \mathcal{C}^{op} & \xrightarrow{\quad} & \mathrm{Fun}(\mathcal{C}, \mathcal{S}) \\ \downarrow & & \downarrow L \\ \mathcal{C}[W^{-1}]^{op} & \xrightarrow{\quad} & \mathrm{Fun}(\mathcal{C}[W^{-1}], \mathcal{S}) \end{array}$$

is commutative, where the horizontal maps are the Yoneda embedding.

Clearly, given any functor $F: \mathcal{C} \rightarrow \mathcal{S}$, we can form a functor $L(F): \mathcal{C} \rightarrow \mathcal{S}$ which sends $X \in \mathcal{C}$ to

$$\mathrm{colim}_{Y \in \mathcal{D}/X} F(\mathrm{cofib}(Y \rightarrow X)).$$

It is easy to check that L is the desired functor. □

Chapter 2

Duality for compactly generated stable categories

Definition 2.0.1. Let $\mathcal{C}$ be a symmetric monoidal category. An object $X \in \mathcal{C}$ is called *dualizable* if there exists another object $X^\vee$ and morphisms

$$\mathrm{ev}: X^\vee \otimes X \rightarrow 1, \quad \mathrm{coev}: 1 \rightarrow X \otimes X^\vee,$$

such that the compositions

$$\begin{aligned} \left[X \xrightarrow{\mathrm{coev} \otimes \mathrm{id}} X \otimes X^\vee \otimes X \xrightarrow{\mathrm{id} \otimes \mathrm{ev}} X \right] &\simeq \left[X \xrightarrow{\mathrm{id}} X \right], \\ \left[X^\vee \xrightarrow{\mathrm{id} \otimes \mathrm{coev}} X^\vee \otimes X \otimes X^\vee \xrightarrow{\mathrm{ev} \otimes \mathrm{id}} X^\vee \right] &\simeq \left[X^\vee \xrightarrow{\mathrm{id}} X^\vee \right]. \end{aligned}$$

In particular, if $\mathcal{C}$ is closed and X is dualizable, then $X^\vee$ must be $\underline{\mathrm{Hom}}(X, 1)$.

This standard compact-generation argument is the part of the continuous K-theory background used here; see [Efi24].

Proposition 2.0.2. For a small stable idempotent-complete category $\mathcal{C}_0$, the category $\mathrm{Ind}(\mathcal{C}_0)$ is dualizable in Pr_{st}^L , with dual $\mathrm{Ind}(\mathcal{C}_0^{op})$.

Proof. Let $\mathcal{C}_0 \in \mathrm{Cat}^{perf}$, we show that $\mathrm{Ind}(\mathcal{C}_0)$ is dualizable with dual $\mathrm{Ind}(\mathcal{C}_0^{op})$. The Ind-extension of the functor $\mathrm{map}_{\mathcal{C}_0}: \mathcal{C}_0^{op} \times \mathcal{C}_0 \rightarrow \mathrm{Sp}$, which preserves finite colimits in each variable, gives an evaluation map

$$\mathrm{ev}: \mathrm{Ind}(\mathcal{C}_0^{op}) \otimes \mathrm{Ind}(\mathcal{C}_0) \rightarrow \mathrm{Sp}.$$

Since $\mathrm{Ind}(\mathcal{C}_0) \otimes \mathrm{Ind}(\mathcal{C}_0^{op})$ is stable, a map $\mathrm{Sp} \rightarrow \mathrm{Ind}(\mathcal{C}_0) \otimes \mathrm{Ind}(\mathcal{C}_0^{op}) \simeq \mathrm{Ind}(\mathcal{C}_0 \otimes^{Rex} \mathcal{C}_0^{op})$ is equivalent to a single object. Recall that a single object of $\mathrm{Ind}(\mathcal{C}_0 \otimes^{Rex} \mathcal{C}_0^{op})$

is equivalent to a finite limit preserving functor $(\mathcal{C}_0 \otimes \mathcal{C}_0^{op})^{op} \rightarrow \mathbf{An}$, i.e. a functor $\mathcal{C}_0^{op} \times \mathcal{C}_0 \rightarrow \mathbf{Sp}$ which preserves finite limit. The coevaluation map $\text{coev}: \mathbf{Sp} \rightarrow \text{Ind}(\mathcal{C}_0) \otimes \text{Ind}(\mathcal{C}_0^{op})$ can be given by the same map $\text{map}_{\mathcal{C}_0}$ and they obviously satisfies the required conditions.

□

Chapter 3

Tate Construction

3.1 Tate construction for finite groups

Definition 3.1.1. Let G be a group, and $\mathcal{C}$ an ∞ -category. An G -equivariant object in $\mathcal{C}$ is a functor $BG \rightarrow \mathcal{C}$, where BG is a fixed classifying space.

The ∞ -category $\mathcal{C}^{BG}$ of G -equivariant objects in $\mathcal{C}$ is the functor ∞ -category $\mathrm{Fun}(BG, \mathcal{C})$.

Definition 3.1.2. Let G be a group and $\mathcal{C}$ an ∞ -category.

1. Assume that $\mathcal{C}$ admits all colimits indexed by BG . The homotopy orbits functor is given by

$$-_{hG}: \mathcal{C}^{BG} \rightarrow \mathcal{C}: (F: BG \rightarrow \mathcal{C}) \mapsto \mathrm{colim}_{BG} F.$$

2. Assume that $\mathcal{C}$ admits all limits indexed by BG . The homotopy fixed points functor is given by

$$-^{hG}: \mathcal{C}^{BG} \rightarrow \mathcal{C}: (F: BG \rightarrow \mathcal{C}) \mapsto \lim_{BG} F.$$

Now assume that G is finite, we will construct a norm map $\mathrm{Nm}_G: -_{hG} \rightarrow -^{hG}$ when the category $\mathcal{C}$ is semiadditive and admits limits and colimits indexed by finite groupoids. [[Lur17, p. 6.1.6.19]]

Construction Let G be a finite group. Then the classifying space BG is a finite groupoid. Let $f: BG \rightarrow \Delta^0$ be the projection map, then the pullback functor $f^*: \mathcal{C}^{\Delta^0} \simeq \mathcal{C} \rightarrow \mathcal{C}^{BG}$ admits left and right adjoints $f_!$ and f_* , given by left and right Kan extension along f . ([Lur09, p. 4.3.3.7]) In particular, we have $f_! = -_{hG}$ and $f_* = -^{hG}$.

Let $\delta: BG \rightarrow BG \times_{\Delta^0} BG$ be the diagonal map, then we have the norm map $\text{Nm}_\delta: \delta_! \rightarrow \delta_*$. Since $\mathcal{C}$ is semiadditive, the natural transformation Nm_δ is an equivalence and admits a homotopy inverse Nm_δ^{-1} .

Let $p_0, p_1: BG \times_{\Delta^0} BG \rightarrow BG$ be the two projections. We have a natural transformation of functors

$$p_0^* \rightarrow \delta_* \delta^* p_0^* \simeq \delta_* \xrightarrow{\text{Nm}^{-1}} \delta_! \simeq \delta_! \delta^* p_1^* \rightarrow p_1^*$$

which is adjoint to a natural transformation $\beta: \text{id}_{\mathcal{C}^{BG}} \rightarrow p_{0*} p_1^*$. The pullback diagram of $BG \times_{\Delta^0} BG$ gives an equivalence between $f^* f_* \rightarrow p_{0*} p_1^*$, so β determines a map $\text{id}_{\mathcal{C}^{BG}} \rightarrow f^* f_*$, which is adjoint to a natural transformation $\text{Nm}_f: f_! \rightarrow f_*$. We call the map Nm_f as the norm map determined by f .

Let M be a G -equivariant object of $\mathcal{C}$, and we identify M with its image in $\mathcal{C}$. There are canonical maps $M \rightarrow M_{hG}$ and $M^{hG} \rightarrow M$. Unwinding the definitions, the composition

$$M \rightarrow M_{hG} \xrightarrow{\text{Nm}_G} M^{hG} \rightarrow M$$

is given by $\sum_{g \in G} \phi_g$, where for each $g \in G$, the map $\phi_g: M \rightarrow M$ is given by evaluation on the 1-simplex of BG .

Definition 3.1.3. Let $\mathcal{C}$ be a stable ∞ -category which admits all limits and colimits indexed by BG for some finite group G . The Tate construction is the functor

$$-^{tG}: \mathcal{C}^{BG} \rightarrow \mathcal{C}: X \mapsto \text{cofib}(\text{Nm}_G: X_{hG} \rightarrow X^{hG}).$$

We will use this definition in particular in the case where $\mathcal{C} = \text{Sp}$. If one takes the Tate spectrum of an Eilenberg-MacLane spectrum HM for a G -module M then one gets back classical Tate cohomology, more precisely $\pi_i(HM^{tG}) \simeq \hat{H}^{-i}(G, M)$.

Assume that G is a normal subgroup of some group H , then for any H -equivariant object $X \in \mathcal{C}^{BH}$, the G -homotopy orbits and G -homotopy fixed points acquire a remaining H/G -action. Consider the projection $f: BH \rightarrow$

$B(H/G)$, then by [Lur17, p. 6.1.6.3] we have the following commutative diagrams

$$\begin{array}{ccc} \mathcal{C}^{BH} & \xrightarrow{f_!} & \mathcal{C}^{B(H/G)} \\ \downarrow & & \downarrow \\ \mathcal{C}^{BG} & \xrightarrow{-_{hG}} & \mathcal{C} \end{array} \quad \begin{array}{ccc} \mathcal{C}^{BH} & \xrightarrow{f_*} & \mathcal{C}^{B(H/G)} \\ \downarrow & & \downarrow \\ \mathcal{C}^{BG} & \xrightarrow{-_{hG}} & \mathcal{C} \end{array}$$

where the vertical functors are the forgetful functors. By abuse of notation, we sometimes denote the functor $f_!, f_*$ by

$$-_{hG}, -^{hG}: \mathcal{C}^{BH} \rightarrow \mathcal{C}^{B(H/G)}.$$

In this case, we still have the Tate construction functor $-^{tG}: \mathcal{C}^{BH} \rightarrow \mathcal{C}^{B(H/G)}$.

3.2 Tate orbit lemma

Lemma 3.2.1. *Let X be a spectrum with a C_{p^2} -action. Assume that X is bounded below, then we have*

$$(X_{hC_p})^{t(C_{p^2}/C_p)} \simeq 0.$$

Lemma 3.2.2. *Let X be a spectrum with a C_{p^2} -action. Assume that X is bounded above, then we have*

$$(X^{hC_p})^{t(C_{p^2}/C_p)} \simeq 0.$$

Bibliography

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